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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('navigation_data.csv')

df.head()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 5000,\n  \"fields\": [\n    {\n      \"column\": \"timestamp\",\n      \"properties\": {\n        \"dtype\": \"object\",\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          \"2023-01-01 00:25:01\",\n          \"2023-01-01 00:43:06\",\n          \"2023-01-01 00:44:13\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"latitude\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.005792673022103504,\n        \"min\": 37.76490023269511,\n        \"max\": 37.78489435346573,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          37.7744836375526,\n          37.766335516124535,\n          37.76594710040375\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"longitude\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.005712330186196142,\n        \"min\": -122.42939894346136,\n        \"max\": -122.40940989620584,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          -122.42779873194942,\n          -122.42105103432206,\n          -122.41315365334572\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"altitude\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 130.84641797819683,\n        \"min\": 50.07098506012553,\n        \"max\": 499.9123148602107,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          448.02344065723855,\n          143.83739313463448,\n          254.7732671636624\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"imu_acc_x\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.726837306876197,\n        \"min\": -2.998485273056628,\n        \"max\": 2.999548960999059,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          1.28373253945231,\n          0.9139152093213054,\n          2.0107569800592247\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"imu_acc_y\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.7324476050522475,\n        \"min\": -2.999711256634129,\n        \"max\": 2.99940586205539,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          -2.219018436733988,\n          1.034689410955428,\n          2.128099248557211\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"imu_acc_z\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.7324476050522475,\n        \"min\": -2.999711256634129,\n        \"max\": 2.99940586205539,\n        \"num_unique_values\": 5000,\n        \"samples\": [\n          -2.219018436733988,\n          1.034689410955428,\n          2.128099248557211\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}"

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```



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```

```

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('UAV Navigation Data Over Time', fontsize=16)

```

```

# Plot Latitude

```

```

axes[0, 0].plot(df_subset['latitude'])
axes[0, 0].set_title('Latitude Over Time')
axes[0, 0].set_ylabel('Latitude')
axes[0, 0].grid(True, linestyle='--', alpha=0.7)

```

```

# Plot Longitude

```

```

axes[0, 1].plot(df_subset['longitude'],color="green")
axes[0, 1].set_title('Longitude Over Time')
axes[0, 1].set_ylabel('Longitude')
axes[0, 1].grid(True, linestyle='--', alpha=0.7)

```

```

# Plot Altitude

```

```

axes[1, 0].plot(df_subset['altitude'])
axes[1, 0].set_title('Altitude Over Time')
axes[1, 0].set_xlabel('Time (seconds)')
axes[1, 0].set_ylabel('Altitude')
axes[1, 0].grid(True, linestyle='--', alpha=0.7)

```

```

# Plot Wind Speed

```

```

axes[1, 1].plot(df_subset['wind_speed'],color="red")
axes[1, 1].set_title('Wind Speed Over Time')
axes[1, 1].set_xlabel('Time (seconds)')
axes[1, 1].set_ylabel('Wind Speed')
axes[1, 1].grid(True, linestyle='--', alpha=0.7)

```

```

plt.tight_layout(rect=[0, 0.03, 1, 0.95]) # Adjust layout to prevent

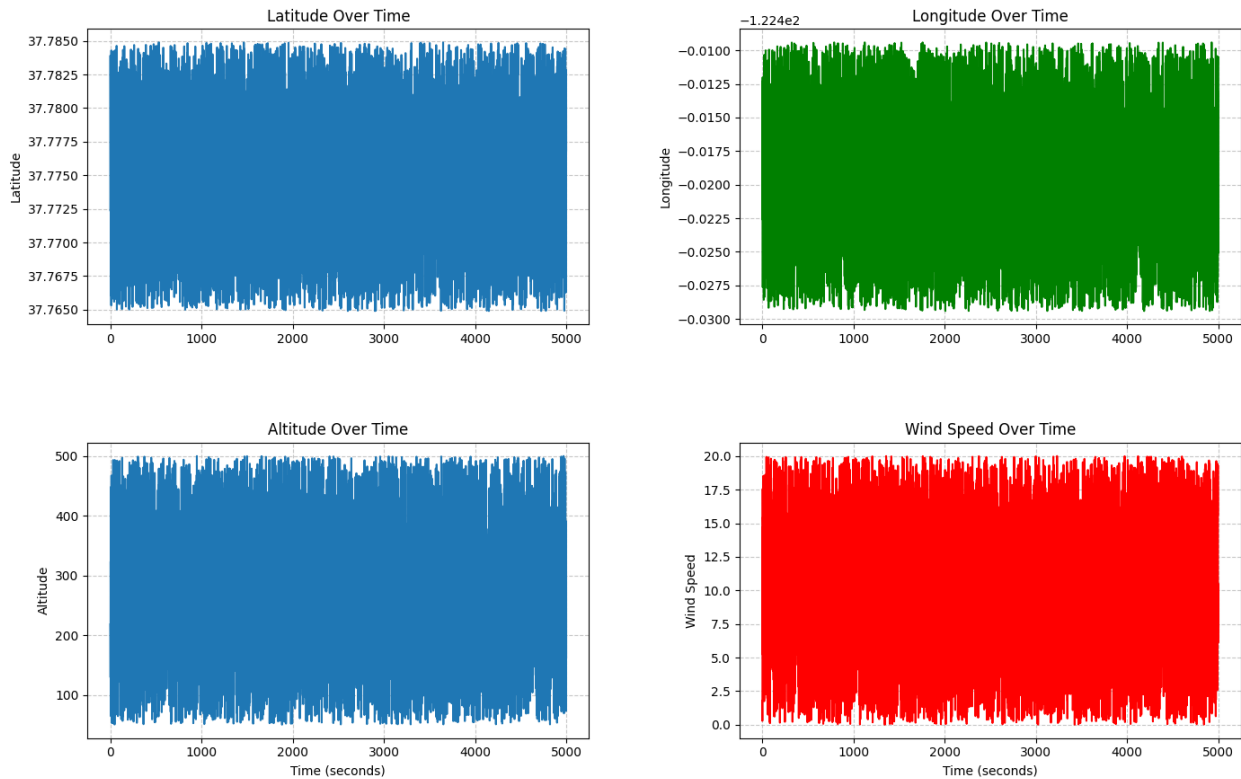
```

```

title overlap
plt.subplots_adjust(wspace=0.3, hspace=0.4) # Adjust padding between
subplots
plt.show()

```

UAV Navigation Data Over Time



Plot Explanations

- **Latitude Over Time:** This plot shows the variation of the UAV's latitude over the recorded time. It indicates the north-south movement of the drone. The plot displays rapid fluctuations, suggesting frequent adjustments or environmental disturbances rather than a smooth, continuous path.
- **Longitude Over Time:** This plot displays the changes in the UAV's longitude throughout the flight, illustrating its east-west movement. Similar to latitude, it exhibits high-frequency oscillations, indicating dynamic movement along the east-west axis.
- **Altitude Over Time:** This graph represents the UAV's altitude (height) at different points in time, revealing its vertical movement and any changes in elevation. The altitude appears to fluctuate significantly within a certain range, possibly due to maintaining a target altitude in varying conditions or sensor noise.
- **Wind Speed Over Time:** This plot illustrates how the wind speed varied during the recorded period, which can impact the UAV's navigation and stability. The wind speed also shows considerable variability, with rapid changes that could influence the drone's flight stability and trajectory.

```

min_lat, max_lat = df_subset['latitude'].min(),
df_subset['latitude'].max()
min_lon, max_lon = df_subset['longitude'].min(),
df_subset['longitude'].max()
min_alt, max_alt = df_subset['altitude'].min(),
df_subset['altitude'].max()
min_wind, max_wind = df_subset['wind_speed'].min(),
df_subset['wind_speed'].max()

# lowest value -10 to highest value +10
lat_lim_low, lat_lim_high = min_lat - 10, max_lat + 10
lon_lim_low, lon_lim_high = min_lon - 10, max_lon + 10
alt_lim_low, alt_lim_high = min_alt - 10, max_alt + 10
wind_lim_low, wind_lim_high = min_wind - 10, max_wind + 10
fig, axes = plt.subplots(1, 3, figsize=(20, 6)) # 1 row, 3 columns
fig.suptitle('UAV Navigation Data with Mixed Axis Limits',
fontsize=16)
# Subplot 1: Latitude vs Longitude Scatter Plot
axes[0].scatter(df_subset['longitude'], df_subset['latitude'], s=5,
alpha=0.5)
axes[0].set_title('Latitude vs Longitude (Scatter)')
axes[0].set_xlabel('Longitude')
axes[0].set_ylabel('Latitude')
# Removed custom xlim and ylim for latitude and longitude
axes[0].grid(True, linestyle='--', alpha=0.7)

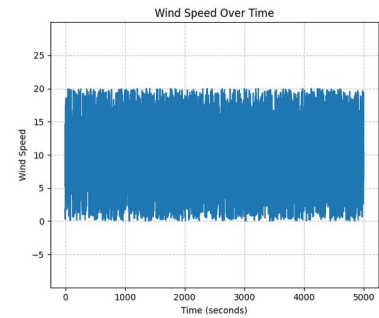
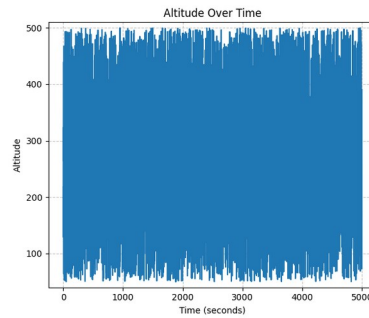
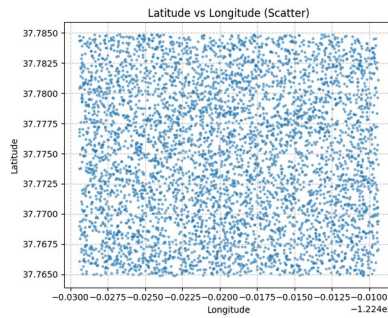
# Subplot 2: Altitude Line Plot
axes[1].plot(df_subset['altitude'])
axes[1].set_title('Altitude Over Time')
axes[1].set_xlabel('Time (seconds)')
axes[1].set_ylabel('Altitude')
axes[1].set_ylim(alt_lim_low, alt_lim_high)
axes[1].grid(True, linestyle='--', alpha=0.7)

# Subplot 3: Wind Speed Line Plot (interpreting 'noise' as wind speed
variability)
axes[2].plot(df_subset['wind_speed'])
axes[2].set_title('Wind Speed Over Time')
axes[2].set_xlabel('Time (seconds)')
axes[2].set_ylabel('Wind Speed')
axes[2].set_ylim(wind_lim_low, wind_lim_high)
axes[2].grid(True, linestyle='--', alpha=0.7)

plt.tight_layout(rect=[0, 0.03, 1, 0.95]) # Adjust layout to prevent
title overlap
plt.subplots_adjust(wspace=0.3, hspace=0.4) # Adjust padding between
subplots
plt.show()

```

UAV Navigation Data with Mixed Axis Limits



```
df_subset.describe()
```

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